

Rohan Pandey

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EDUCATION

University of Washington, Seattle, WA

B.S. March 2026

B.S., Applied & Computational Mathematical Sciences: Scientific Computing & Numerical Analysis • Returning Fall 2026 for

M.S. in ACMS • Minor: Neural Computation & Engineering

Relevant Coursework: Machine / Reinforcement Learning, High-Performance Scientific Computing, Optimization, Data Structures & Algorithms, Database Systems

Awards: Lead AI Researcher Award, UW Math AI Lab • NASA Space Grant Awardee

EXPERIENCE

Software Engineer II, DigitalOcean (Dedicated Inference)

March 2026 – Present

- Own inference benchmarking for the dedicated GPU fleet: orchestrate **vLLM on Kubernetes** to measure **throughput and latency** across heterogeneous GPU hardware, model configurations, and batching / concurrency settings; results set capacity, routing, and SLA decisions for production inference workloads
- Build **prefill-decode (PD) disaggregation** and **KV cache reuse with LMCache**: split prefill and decode across GPU pools and share cached prefixes to cut time-to-first-token and raise sustained throughput on production inference paths
- Built and shipped a **Kernel Optimization Agent** that profiles and auto-tunes **GPU kernels** using **Optuna**-driven search over kernel configurations, with automated benchmarking in the loop to validate speedups on real inference paths
- Debug **throughput and latency regressions** end-to-end across the GPU, Linux, and networking stack; own production observability for the fleet (Prometheus, Grafana, PagerDuty)

Lead Reinforcement Learning Researcher, UW Math AI Lab

September 2025 – Present

- Built a **PPO + Graph Neural Network** agent that synthesizes minimal arithmetic circuits, reaching **~70% success on degree-3 polynomials** via curriculum learning, shaped rewards, and symbolic verification; scaled with **MCTS + AlphaZero-style self-play**; **first author at ICML 2026 and ICLR 2026**
- Lead the RL sub-team: own experiment design, the evaluation harness, and the open-source benchmark release, and the profiling that kept training runs cheap enough to iterate daily

Machine Learning Engineer, Mercor

December 2025 – March 2026

- Built and deployed an end-to-end ML pipeline for text sentiment analysis at scale; fine-tuned and evaluated transformer models with custom scoring metrics

Cloud & Digital SAP Intern, PricewaterhouseCoopers

June – August 2025

- Engineered high-throughput Python data pipelines (vectorization, parallel I/O) to cut ingestion latency; shipped a secure self-service web app for configuring and monitoring automated workflows

SELECTED PUBLICATIONS

14 papers and preprints, 3 at ICML / ICLR 2026 workshops. Full list at [publications site](#) • [Google Scholar](#)

- [Quantization Blindspots: How Model Compression Breaks Backdoor Defenses](#) • arXiv:2512.06243: INT8 quantization drops backdoor-detection rates to 0% while attack success stays above 99%
- [ThinkSwitch: Context Distillation with LoRA and Weight Interpolation for Reasoning Tasks](#) • arXiv:2606.01080: low-compute co-training of paired instruct / thinking Qwen3-4B checkpoints
- [FactorLibrary: From Polynomials to Circuits via Recursive Subgoals](#) • ICML 2026, AI for Math Workshop (first author)

PROJECTS & AWARDS

- [gauntlet-agent](#) (open source, PyPI • MIT): adversarial stress-testing harness for AI agents, with deterministic probes for prompt injection, unsafe tool calls, scope drift, and infinite loops; framework-agnostic over HTTP, with severity-ranked findings, multi-turn jailbreak probes, and CI/CD exit codes
- **Hackathon wins:** Winner, *Husky Maps* (UW Claude Hackathon 2025); 2nd Place, *BloomSync AI* (DubHacks); Top Project, *NeuralTrace* (NeuroAI Hackathon, real-time EEG visualizer)

SKILLS

Inference & GPU: vLLM, LMCache, CUDA, Optuna, prefill-decode disaggregation, KV cache reuse, GPU profiling and benchmarking, throughput / latency / concurrency tuning, INT8 quantization, LoRA, PyTorch, Transformers, performance debugging

Systems & Infra: Linux, Kubernetes, Docker, Prometheus, Grafana, PagerDuty, networking, FastAPI, PostgreSQL, Git / GitHub Actions, DigitalOcean

Languages: Python, C++, Java, SQL, Bash, Lean, MATLAB

ML / Research: Fine-tuning & context distillation, RL (PPO, MCTS, AlphaZero), GNNs, LLM agents and tool-use, evaluation design